ALGEBRAIC NUMBER THEORY
MATHEMATICS GU4043
SPRING 2026

Instructor: Gyujin Oh (gyujinoh@math.columbia.edu), pronouns: he/him/his.
Webpage: https://math.columbia.edu/∼gyujinoh/∼Spring2026.html, and Courseworks.
Time and location: MW 4:10-5:25PM, at 652 Schermerhorn Hall.
Office hours: Wednesdays 3-4PM, at 517 Mathematics.
Teaching assistants: Analisa Faulkner Valiente (acf2182@barnard.edu), pronouns: they/them/theirs.
TA Office hours: Fridays 4-6PM, at 406 Mathematics.

Textbook: An Invitation To Modern Algebraic Number Theory (`https://www.math.columbia.edu/~gyujinoh/ANT_revised.pdf`)

Prerequisites: MATH GU4041 and MATH GU4042 or the equivalent. This includes: Groups, homomorphisms, normal subgroups, the isomorphism theorems, symmetric groups, group actions, the Sylow theorems, finitely generated abelian groups, rings, homomorphisms, ideals, integral and Euclidean domains, the division algorithm, principal ideal and unique factorization domains, fields, algebraic and transcendental extensions, splitting fields, finite fields, Galois theory.

In the latter part of the course, when we learn about zeta functions and L-functions, we will use elementary complex analysis (mostly the notion of holomorphic functions and contour integrals). It will be helpful (but not necessary) to know the basics of complex analysis.

What this course is about: This course is an introduction to **algebraic number theory**. The following list is more or less the complete list of topics covered in a standard first course in algebraic number theory, which I will cover in order.

- Basic commutative algebra.
- Number fields and their rings of integers.
- Ramification theory.
- Dirichlet's unit theorem.
- Finiteness of class number.
- Local fields.

I will also try to incorporate some hints to more modern developments of algebraic number theory, such as the Langlands program or the Birch–Swinnerton-Dyer conjecture, via their precursors found in class field theory and cyclotomic fields. The more advanced topics include:

- How to understand the local and global class field theories (without proving them).
- Applications of the class field theory.
- The arithmetic significance of L-functions and their special values.
- Cyclotomic fields.

Grading: There will be weekly homework assignments (45%), Midterm (20%), Final (20%), and an individual 15-minute presentation (15%). **The lowest homework will be dropped.**

The first half of the homework assignments (HW1∼6) are due on Mondays, with HW1 due by Monday, February 2. This schedule changes after the Spring Recess, where the second half of the homework assignments (HW7∼12) will be due on Wednesdays, with an exception that HW12 is due by the last day of classes, Monday, May 4. More precisely, the schedule is as follows.

- HW1: by Monday, February 2.
- HW2: by Monday, February 9.

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- HW6: by Monday, March 9 (the Monday before the Spring Recess).
  - The Midterm is on Wednesday, March 11.
- HW7: by **Wednesday, March 25** (the Wednesday after the Spring Recess).
- HW8: by Wednesday, April 1.

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- HW11: by Wednesday, April 22.
- HW12: by **Monday, May 4** (the last day of classes).

Both exams will be held in person. The Midterm will be held in-class on **Wednesday, March 11**. The Final is projected to be held on **Monday, May 11 at 4:10-7pm**. Please let the instructor know as early as possible if you think you will not be able to take the in-class exam on that date.

You will be asked to give a 15-minute presentation on anything a step further than an advanced topic taught in this course (e.g., class field theory, arithmetic of cyclotomic fields). Most likely you will choose a part of a paper from a list that I will suggest, but you are also welcome to come up with your own. The date of the presentation is TBD.

Accessibility and accommodations: Your success in this class is very important. We all learn differently. If there are aspects of this course that prevent you from learning or exclude you, please let me know as soon as possible. We can develop strategies to meet both your needs and the requirements of the course.

If you think you might need official accommodations, such as extended time on exams, I encourage you to contact the Office of Disability Services (Columbia) or CARDS (Barnard) for a confidential discussion. Once you register with them, they can provide you with an accommodation letter, which will allow you to receive official accommodations.

Staying well: Math, and college, can be hard. If you are facing challenges related to your physical or mental health, or any sort of difficulties, you are encouraged to contact your advising dean and/or the Student Health Service. If you feel comfortable doing so, please do not hesitate to get in touch with me to discuss ways we can put you in the best possible position to succeed. If you're finding yourself overwhelmed but don't get help, then the tide may very well sweep you away and leave you completely lost!

Inclusivity: We are part of a learning community and must treat one another with respect at all times. This is especially important with regard to race, religion, nationality, sexual orientation, gender, disability, age, size, immigration status, parental status, and any other aspect of identity. I am committed to ensuring that this class is a supportive, inclusive, and safe environment for all students, and that all students are treated with dignity and respect. See also the Columbia College Notice of Non-Discrimination.